M Navya Sree

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EDUCATION

Vellore Institute of Technology 2021 – Present
Integrated M.Tech in Computer Science and Engineering *CGPA: 8.06*
Mangal Vidyalayam 2019 – 2021
Central Board of Secondary Education

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS
Software Engineering: Software Development Life Cycle (SDLC), Agile, Scrum, Requirement Analysis, Software Testing, Debugging, Version Control, Code Review
Machine Learning: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, CNN, LSTM, NLP, Data Preprocessing, Feature Engineering, Model Evaluation
Generative AI: Prompt Engineering, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, AI Agents,Artificial Intelligence
Cloud & DevOps: Microsoft Azure Fundamentals, Docker, CI/CD Fundamentals
Databases: MySQL, SQL, Database Management System
Tools: Git, GitHub, Linux, Jupyter Notebook, VS Code

EXPERIENCE

Technology Intern Aug 2025 – Jun 2026
Virtusa

- Contributed to designing and building modular backend workflows for an enterprise Property Management System, working with NestJS modules, controllers, providers, and dependency injection patterns.
- Helped develop and integrate RESTful APIs connecting NestJS/Spring Boot backend services with the Angular frontend, implementing JWT-based authentication for secure, role-based access.
- Applied Agile engineering practices – sprint planning, code reviews, Git/PR workflows – to improve maintainability and streamline development cycles across multiple modules.
- Collaborated closely with cross-functional teams on API integration and feature delivery, contributing to production-oriented enterprise application modules.

PROJECTS

Stock Market Trend Forecasting using CNN–LSTM | Python, TensorFlow, Deep Learning
- Developed a hybrid CNN–LSTM deep learning model for time-series stock prediction using historical financial datasets.
- Built an end-to-end pipeline including data preprocessing, feature engineering, sequence generation, and model training.
- Achieved R^2 score of 0.97, with RMSE of 5.03 and MAE of 3.80 demonstrating strong predictive accuracy.
- Combined CNN layers for local pattern extraction with LSTM layers for sequential dependency modeling, improving forecast stability over standalone architectures.
- Tuned hyperparameters (window size, learning rate, batch size) to reduce overfitting and improve generalization on unseen data.

Augmented Reality 3D Jet Turbine Simulation | Unity, AR Foundation
- Developed an interactive AR application enabling visualization of a 3D jet turbine model in real environments.
- Implemented component-level interactions allowing users to rotate, zoom, and isolate turbine components.
- Integrated AR Foundation enabling cross-platform compatibility across Android and iOS devices.
- Designed intuitive touch-based gesture controls to make complex mechanical structures easier to explore and understand.
- Optimized 3D model rendering for smooth real-time performance on mobile AR hardware.

PUBLICATION

Learning Outcomes of Classroom Research Dec 2021
L'Ordine Nuovo Publication – Research contribution on AI applications in education.

CERTIFICATIONS

- Microsoft Azure AI Fundamentals (AI-900)
- Advanced Software Engineering Job Simulation – Walmart Global Tech
- Infosys Springboard – Programming Using Java